



Supriyo Paul

SENIOR SOFTWARE ENGINEER

CONTACT

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EDUCATION

2013-2017
**BACHELOR OF TECHNOLOGY,
COMPUTER SCIENCE**
Mallabhum Institute of Technology

2011-2013
HIGH SCHOOL
D.A.V Public School

2000-2011
SCHOOL
De Nobili School

CERTIFICATES

AI for Everyone, Deeplearning.ai
coursera.org/account/accomplishments/certificate/JMQWSQBKD
SZC

Protocol Buffers, Udemy
udemy.com/certificate/UC-1dd3f3c2-d8ad-46b5-9a09-801f26cf0
df6

Data Structures & Algorithms, Udemy
udemy.com/certificate/UC-d119c94a-5051-48e0-8f9b-6298c8975
b81

System Design, Udemy
udemy.com/certificate/UC-a6688b24-e5c2-48e2-9c10-231f972ea
e68

LeetCode Top 6.9% Badge, 50-Day Challenge

SUMMARY

- Senior Software Engineer with **9+ years** of experience in Python development, system design, and end-to-end automation engineering across enterprise-scale platforms.
- Built and maintained **integration platforms processing 45M+ events** and **79M+ API requests annually** at **98.97% success rate** on GCP (Cloud Run, GKE, Spanner).
- Architected **knowledge-graph microservices** and **ELK/Kafka clusters processing 7B+ records daily** across **16 microservices**.

EXPERIENCE



H&M

Mar 2024-Present

SOFTWARE ENGINEER II

- Developing an **integration platform** with **15+ microservices** handling complex business objects across PLM software and ERPs, serving **10+ downstream systems** via configurable **no-code workflows** on **GCP (Cloud Run, GKE, Spanner)**.
- Contributed to a platform processing **45M+ events** and **79M+ API requests annually** with **98.97% success rate**; supported peak loads of **2.3M+ requests/day** with zero instability.
- Delivered **10+ end-to-end business object integrations**; reduced incident resolution time by **60%** through automated diagnostic scripts and structured debugging runbooks.



Jio Platforms Private Limited

Sep 2021- Mar 2024

SOFTWARE ENGINEER II

- Engineered a **Knowledge-graph platform** with **3 microservices** (Schema, Data, Search) on **ArangoDB** with **Redis** caching, using **gRPC/Protocol Buffers** for inter-service communication.
- Led **performance monitoring and scaling** of Knowledge Graph services on **Kubernetes** with CI/CD pipelines on **Azure DevOps**, ensuring optimal scalability for Big Data ingestion and retrieval.
- Mentored the AICOE AI Platform team, establishing best practices for code quality and maintainability across the ML/MLOps platform.



Ajio (Reliance Retail)

Sep 2019 - Sep 2021

DEVOPS ENGINEER

- Built a **centralized visibility platform** on **ELK and Kafka clusters** processing **7B+ log records daily** across **16 microservices**, pinpointing performance bottlenecks in real-time.
- Led infrastructure benchmarking and load testing for migration to Jio's in-house cloud (JAWS); managed deployments via **Ansible** with **Prometheus** monitoring and **Grafana/Kibana** dashboards.
- Founded the **Tools & Automation team**, implementing strategic automation solutions that streamlined workflows and eliminated redundant tasks across multiple departments.



Deepcompute LLC (nference)

Aug 2017 - Sep 2019

ASSOCIATE SOFTWARE ENGINEER

- Developed **open-source libraries** (logagg, basescript, deeputil) and managed **data pipelines** ingesting large-volume JSON datasets for drug-discovery analytics serving pharma clients.
- Established CI/CD pipelines using **Docker and Travis CI**; enforced code quality benchmarks with automated testing, managing infrastructure on **MongoDB, InfluxDB, and NSQ**.

SIDE PROJECTS



applibot.io

Sep 2023 - Present

CREATOR & SOLE DEVELOPER

- Built an AI-powered **Retrieval Augmented Generation (RAG)** platform to automate job applications (cover letters, ATS resumes, form fill-ups) using **FastAPI, LangChain, React, and Chrome Extension**.
- Designed vector similarity search using **PostgreSQL + pgvector** and **Maximal Marginal Relevance (MMR)** for diverse, non-redundant LLM context retrieval.
- Migrated infrastructure to **GCP (Cloud Run, Firebase)** for \$0/month serverless hosting and implemented professional PDF generation via **WeasyPrint and LaTeX**.

SKILLS

CONCEPTS

System Design • Microservices • Event-Driven Architecture • LLM/RAG • Product Design • ETL Pipelines • gRPC/Protocol Buffers • REST APIs • CI/CD • Knowledge Modeling

LANGUAGES

Python • SQL • Bash • HTML/CSS

FRAMEWORKS

FastAPI • LangChain

DATABASES

Cloud Spanner • PostgreSQL (pgvector) • LanceDB • ArangoDB • Elasticsearch • MongoDB • InfluxDB • Redis

DEVOPS & CLOUD

GCP (Cloud Run, GKE, Pub/Sub, Cloud Build, Firebase) • Kubernetes • Docker • Kafka • Azure DevOps • Ansible • ELK Stack • Prometheus • Grafana • Linux